Maximilian Harkins

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EDUCATION

University of Rochester **Rochester, NY**
Bachelor of Science, Biomedical Engineering, Concentration in Microfluidic Device Eng. *Anticipated May 2027*

- GPA: 3.76 out of 4.00 • Schwartz Discover Grant Recipient (2024)

RESEARCH & EXPERIENCE

Meta Reality Labs × NeuroMechatronics Lab, Carnegie Mellon University **Pittsburgh, PA**
Research Assistant *May 2026 – Aug. 2026*

- Worked under a Meta Reality Labs grant with Meta engineers and Douglas Weber's NeuroMechatronics Lab on EMG-wearable interfaces for computer access in cervical spinal cord injury (cSCI).
- Introduced and implemented a hybrid air-mouse + EMG-click interface driven entirely by a standalone Meta Neural Band; pilot data (n = 3, able-bodied) showed Fitts' throughput rise from 2.3 bits/s (trackball) to 3.6 bits/s (air mouse), vs. 4.4 bits/s with a traditional mouse.
- Built a cross-platform macOS / iOS / Android app for pairing, streaming, & gesture control of the standalone Neural Band.

Zhang Lab – Center for Musculoskeletal Research (CMSR) @URMC **Rochester, NY**
Project Leader *Oct. 2023 – Present*

- Engineering pre-vascularized bone tissue implants in collagen hydrogels to improve bone defect repair.
- Led an interdepartmental collaboration between CMSR and BME to advance 3D optical ablation and perfusion.

BROWN LAB – Department of Biomedical Engineering (BME) @URMC **Rochester, NY**
Project Leader *Dec. 2023 – Present*

- Designed and built the lab's two-photon micromachining platform (WayPoint, below), now its standard tool for 3D controlled ablation; mentored a lab member on it.
- Fabricate microfluidic perfusion devices and cast collagen gels for ablation targets and endothelial seeding.

Department of Undergraduate Research – Research Ambassador **Apr. 2025 – Present**

- Advise students entering undergraduate research; lead discussion tables between ambassadors across all four years.

SELECTED PROJECTS

WayPoint: Two-Photon 3D Micromachining Platform (MATLAB) *Nov. 2024 – Mar. 2026*

- Built and documented a MATLAB application converting arbitrary 3D geometry into executable two-photon scan paths at 800 nm beam-waist sampling – subtractive ablation of in-vivo-derived capillary networks, and additive UV-resin micro-3D printing.
- Wrote the whole stack: a vectorized STL slicer and galvanometer-travel path planner, a zoom-independent µm-to-scan-angle calibration, and NI-DAQmx waveform playback with laser power gated to motion.

Wireless Protocol Reverse Engineering (Personal) *2026*

- Reverse-engineered [redacted] BLE protocols on a personally owned [redacted] from a decompiled, obfuscated Android client, recovering raw EMG telemetry the consumer SDK does not expose.
- Decoded a Backfire G3 electric longboard's proprietary 2.4 GHz protocol byte-by-byte with a serial analyzer; built nRF52840 (Seeed XIAO) bridge firmware and an Android app for full control – throttle, brake, lights, direction.

Standalone Electrooculogram (EOG) Biosignal Amplifier *Apr. 2025 – May 2025*

- Designed, prototyped, and calibrated a $10 standalone EOG amplifier; 1 of 3 teams out of 20+ meeting all performance, cost, time, and safety specs.

PRESENTATION HIGHLIGHTS

- Invited Presenter, Hajim National Council (Nov. 2024) – selected from 100+ students; presented a microfluidic perfusion chamber for repeatable ablation and endothelial seeding.
- Research Presenter, Meta Reality Labs & CMU NeuroMechatronics Lab (Aug. 2026)
- Poster Presenter, CMSR Symposium (Oct. 2025) – "Multiphoton Ablation & Laser-Guided Angiogenesis," department-wide.

SKILLS

Languages & Tools: Python, MATLAB (App Designer, OOP), C/C++ (embedded), Git, LabVIEW, SolidWorks, AutoCAD

Systems & Hardware: BLE / 2.4 GHz protocol reverse engineering, nRF52840 firmware, NI-DAQmx analog I/O

Signal Processing: EMG gesture decoding, MUAP decomposition, IMU sensor fusion & quaternion drift correction, Fitts' law analysis

Wet Lab: Tissue engineering, collagen hydrogels, microfluidic device design, two-photon microscopy

EXTRA-CURRICULAR ACTIVITIES

University of Rochester, Quantitative Finance Club – Vice-President **Feb. 2026 – Present**

- Lead weekly sessions on trading strategy; run an open project track for members' quantitative models.